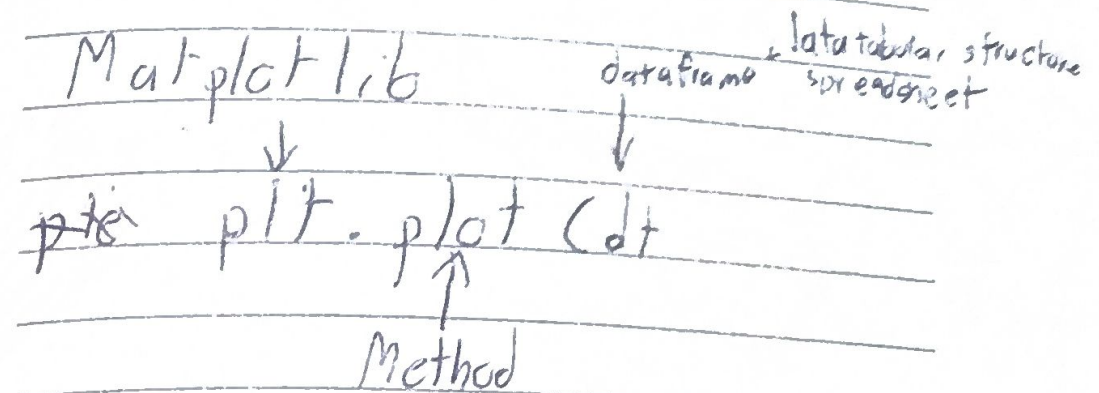


Data Science Essentials with Python

Matplotlib's Pyplot



iloc



accessor

~~access~~ This gets a row from a dataframe

pd = Pandas = contract

Eval = Can use formula new columns

Query can use condition

~~groupby()~~ method groups rows that share a common value

reset_index = volverlo poner formato dataframe

plt

scatter - puntos

bar = barras

barh = columnas

plot = lineas

plt, axvline()

plt, axhline

number

head() is the number they will give from up to down

split = str.split and as a separated data 3:52 7:52

df.info()

split_duration = df['duration'].str.split(' ', expand = true)

Import pandas as pd

df = pd.read_csv('top-song-durations.csv')

df.head(5)

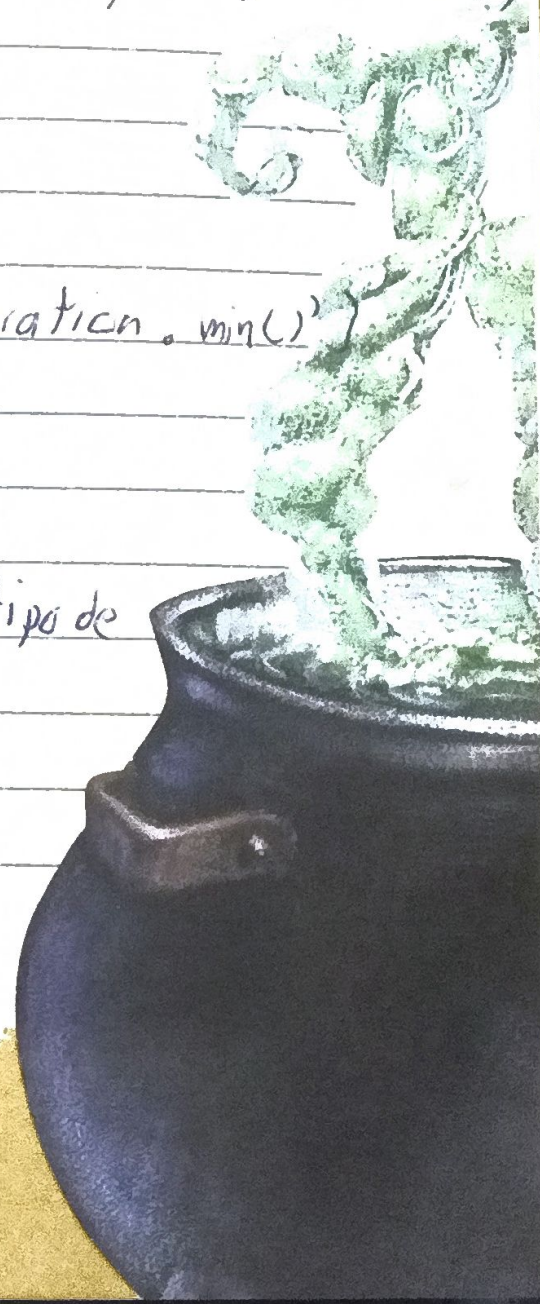
df.query('duration == duration.min()')

string = '00'

astype = modifica tipo de data

eval()

df[] = df.eval



↓
df = df.rename(columns = " ")
df.isna()

This checks if values are missing,
returning true for missing and false
otherwise



↓
df.dropna() = this removes rows
with missing values

mean = promedio

pd.to_datetime() = change
date to date

strftime() = time string

divers['category'].value_counts()

df.loc[Label]

df.loc = ~~df~~.add row

plt.scatter (post_recovery ['shoulder-height'], post_recovery ['tusk-length'], marker = 's')

plt.axhline (y) = add a horizontal line to a plot

plt.axvline (x) = add a vertical

Quadratic Model R^2 = le da congruencia a la linea del centro, la tendencia del grafico

Quadratic plot_model (G, 40)

1 2 3 4 5 7 8
B D B A A A
A C C C A B
C D D B C



gxvspan () = Crear una seccion tipo sombrero en el grafico



Data Analytics in Action

Asking the Question: Encontrando la meta principal para sacar datos

Getting the data: Process of locating and obtaining data, and if there is enough data

Investigating the data: Determining if the data is complete and contains the relevant for the analysis

Preparing the data: Cleaning

1 2 3 4 5 6 7 8
A C C C A B
C D D B C

1 5 6 7 8
4 3 4 3 1 1
25 1
23

D D



42
54
66

Analyzing the data: Analysis is the process of identifying patterns, correlations and relationships between data.

Presenting the results: communicating the results to decision-makers.

Descriptive Analytics: Observed data to identify key characteristics of a data set. It relies solely on historical data to provide reports on past events.
Graphics

Predictive Analytics: Real-time and/or past data to make predictions based on probabilities. It can also be used to infer missing data or establish a predicted future trend. Predictive analytics uses simulation models and forecasting.

Prescriptive Analytics: Builds on descriptive and predictive analysis by recommending courses of action that will reap the greatest benefit for the organization.

Diagnostic Analytics: ~~Diagnostic~~ Descriptive analytics process by digging deeper and attempting to discover the cause.



Quantitative
Continuous: 1.2, 3.4 $\neq$

discrete: 1, 2, 3 $\neq$

Qualitative

nominal: Gender: Sobre the object

ordinal: Education level: clasificación
del objeto in order

Data type

String: Text

Integer: Todo numero entero positivo y negativo

Floating point: Numeros con decimales

Date and time: 12/10/2012 11:53 AM

Boolean: True and false

Streaming data is analyzed and the outcome recorded before it is store or used. Come from lived data (real-time)

Relational Databases: Relation entre los datos

Machine-generated historical record of all of the systems

Spreadsheets: flat file databases stores records

Sensor readings: Sensor-delectura
Transactional Records: can be store of Records in many different formats

ETL : Extract, Transform and Load

ELT : Extract, Load, Transform

Excel : Trim = quita espacios
Proper = deja primera letra en mayuscula
el resto en minuscula de cada
palabra

LEN = Sirve para contar el numero
total de caracteres en una celda

Descriptive Statistics : Summarize data
them so they make sense

Mean, Median, Mode, range, Standard Deviation

Inferential statistics ^{to} pull data
and make smart guesses

Analysis of big data

Cluster analysis: find groups that are
similar

Association analysis: Un analysis
de asociacion. If you buy item A, you will
probably buy item B

Regression analysis: Correlacion
de datos si pasa algo con
A pasa algo en B but in
a quantifi matter

Business analysis

Data engineer

Data analysis

Data Science

~~Los~~ Datos estructurados:

Tiene filas y columnas

Son datos organizados

Datos semi estructurados: no encaja en la Tablas rígidas filas y columnas pero que utiliza etiquetas o metadatos para organizar elementos y jerarquías

Datos no-estructurados: video, correos, fotos son como un caos desordenado

Quien hizo que

como ~~el~~ evoluciona la venta

pronóstico por temporada

Cuanto vendieron

Venta gerente contabilidad

Datos operacionales: son datos son una suma máxima media promedio mínimo. Es todo lo que se puede medir con una regla o un calculadora

1. Clasificar como es

2. Número de hijos color 2 peso 1 color

2 edad 2 peso 1 como 2 fecha

1 hijo 1 y 2 altura

1 cualitativo

2 cuantitativo

Vala separados por comas
(3V)

No guarda imágenes, etc.

la imagen se puede guardar
con un enlace

Taxon Tson: formato de texto ligero
diseñada para el intercambio y almacenamiento de datos
int: son números enteros

Am)

Filas

columnas

llaves foráneas

llaves relacionales

SQL para empresas

